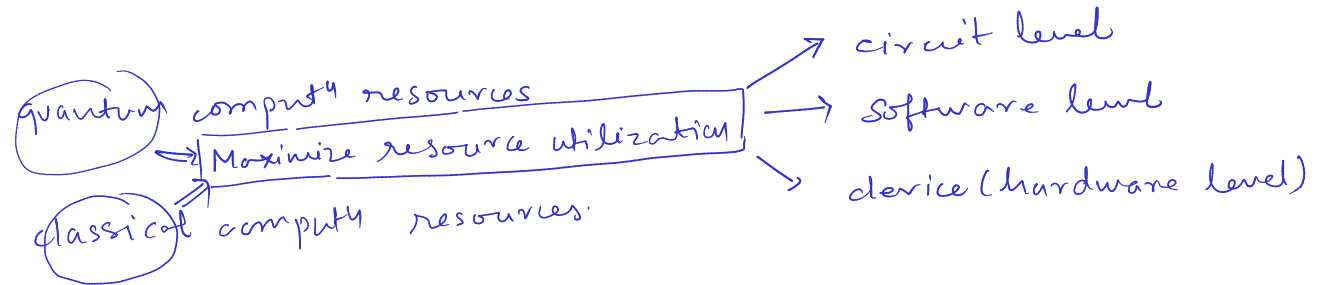


[VQE, QAOA, VQSim, VQFactor, VQML]...

Hybrid Quantum-Classical Computing

“Best of the two worlds” approach to computing

Lakshya Priyadarshi



Q π _{AI} Hybrid Strategy I : Circuit Decompositions



1. **Compile** a quantum algorithm into a quantum circuit

2. **Partitioning** of the circuit into suitable sub-circuits

- Minimize the number of gates connecting the subcircuits
- Subcircuits should fit on current-generation NISQ hardware
- Identify subcircuits that can be simulated efficiently $\rightarrow$ *Quantum gates*
- Simulate error-sensitive qubits on reliable classical hardware

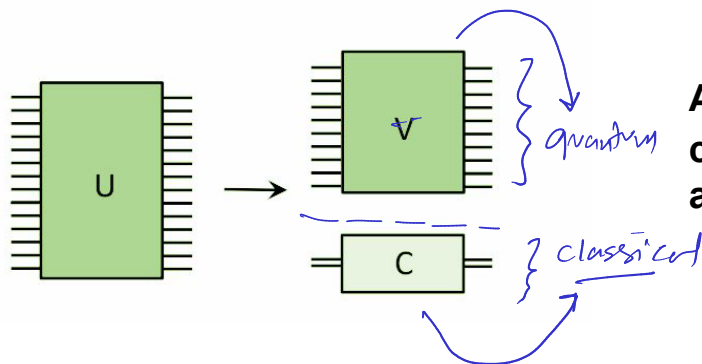
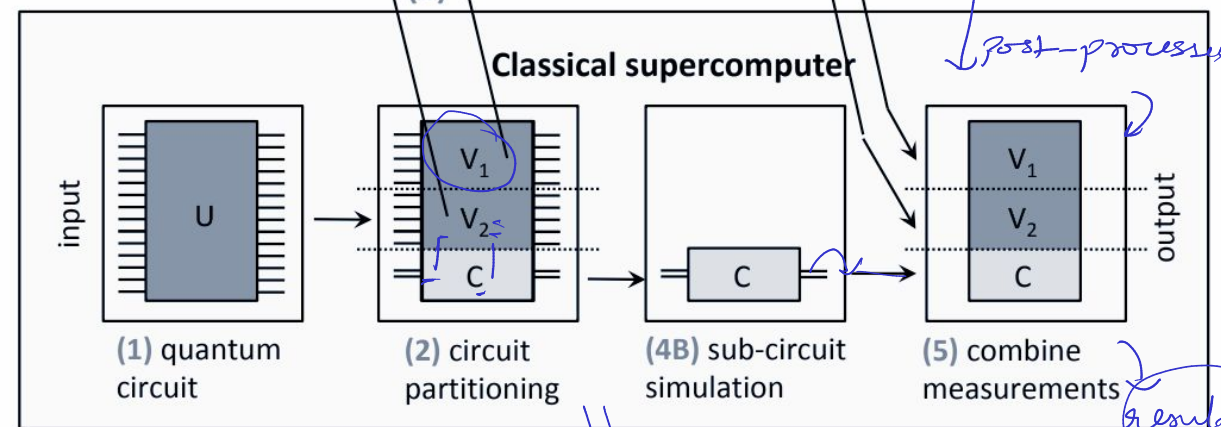
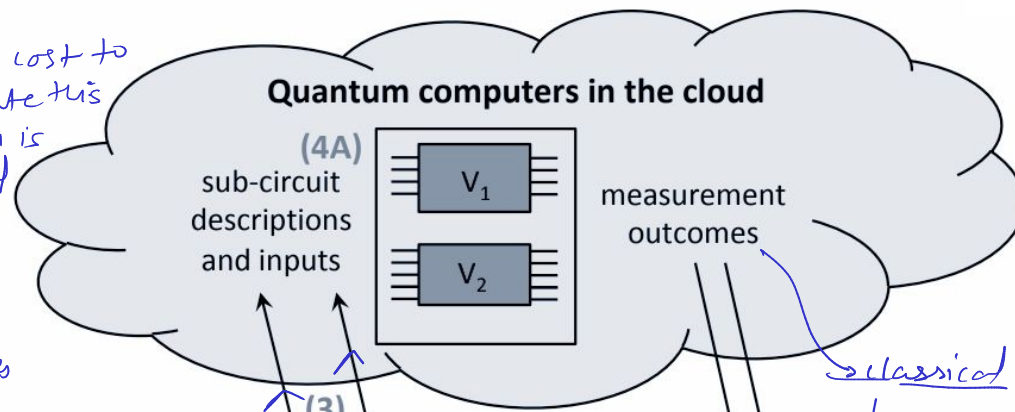
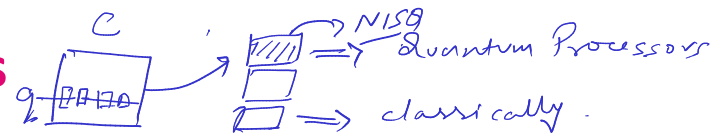
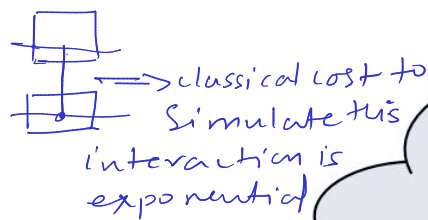
3. **Transfer** circuit descriptions and inputs to quantum processor

- Error-mitigation or error-correction

4. **Computing** on quantum processors and classical processors

- classical processor: classical and quantum circuits

5. **Combining** the results from quantum and classical processors



A quantum computation requiring n qubits can be completed using $n-k$ qubits and additional classical computation resources

REF: Hybrid Quantum-Classical Computing Architectures, Martin Suchara, et. al.

Hybrid Strategy I : Circuit Decompositions

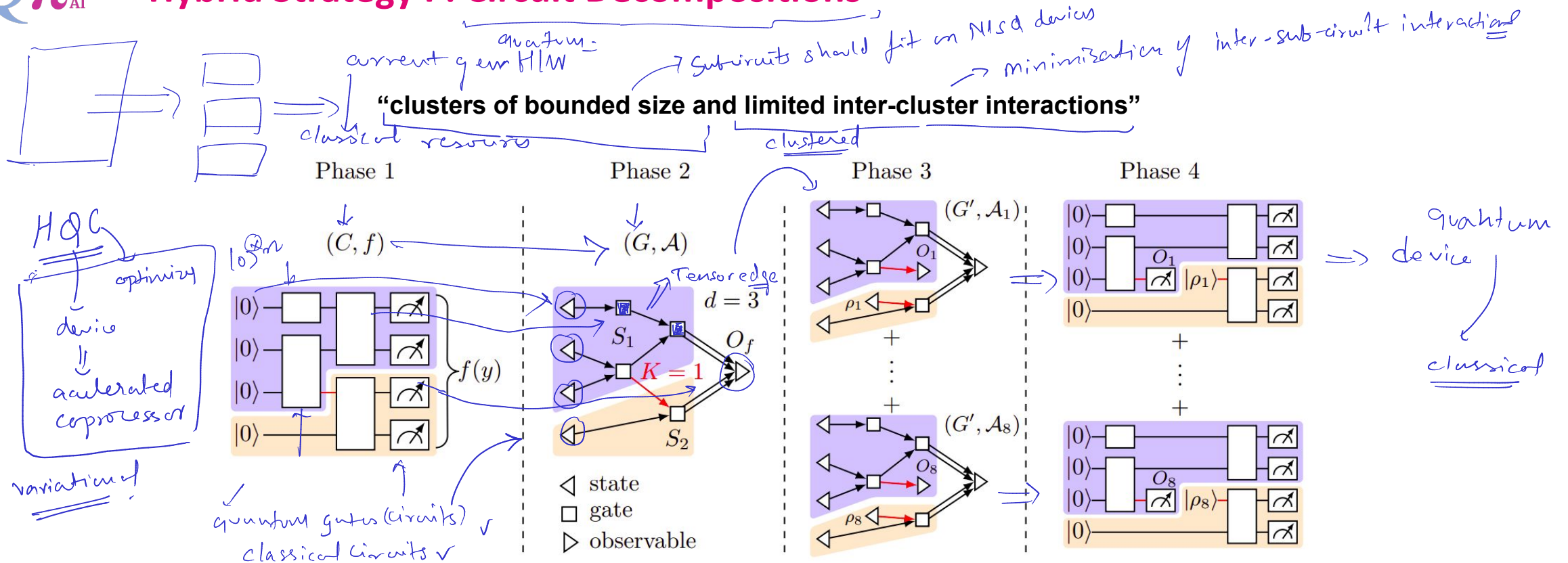
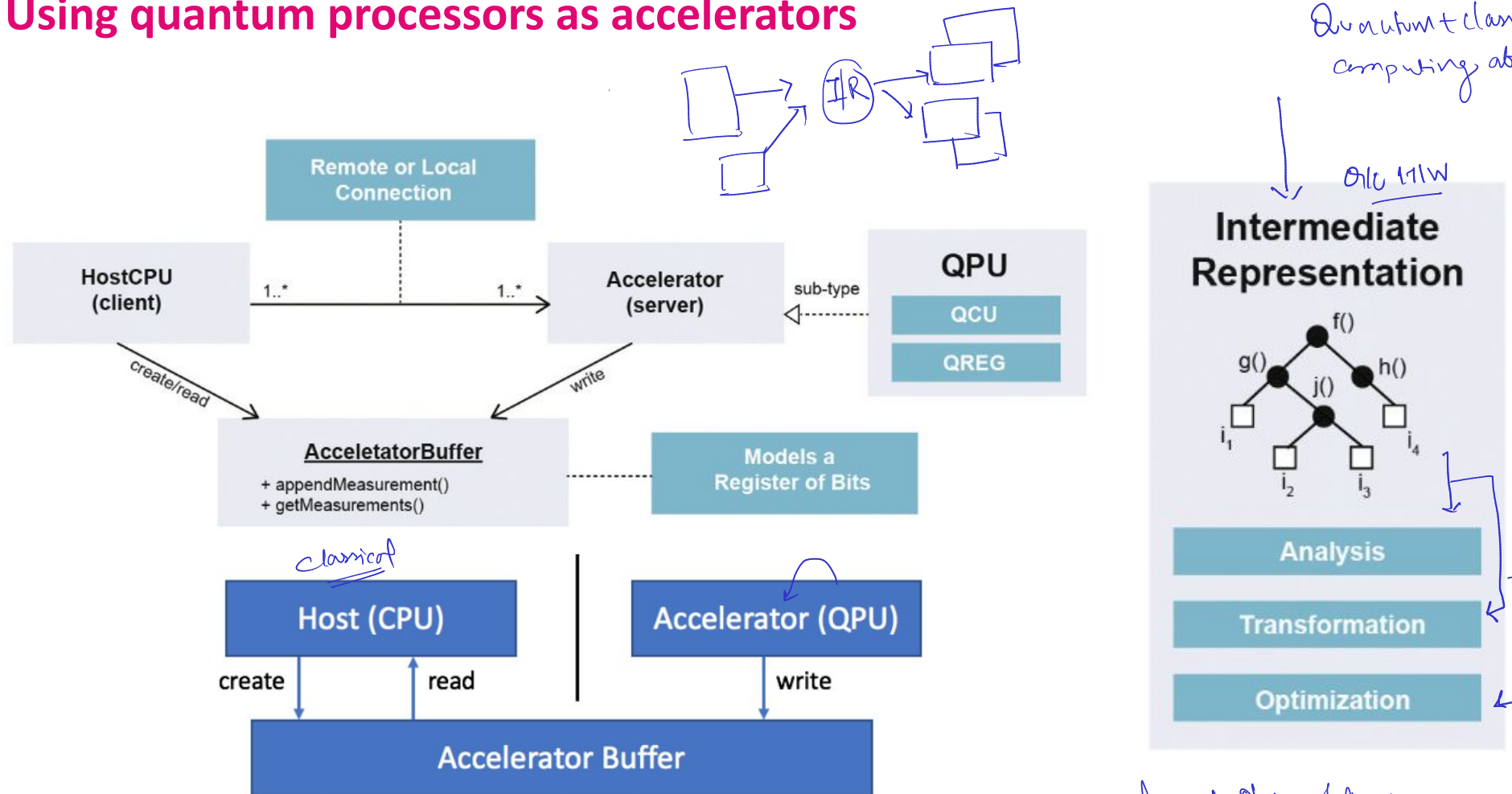


FIG. 1. Four phases of our simulation: (1) the original quantum circuit; (2) the corresponding tensor network; (3) a collection of tensor networks obtained by cutting an edge; (4) a collection of smaller quantum circuits.

REF: Simulating Large Quantum Circuits on a Small Quantum Computer, Tianyi Peng et. al.

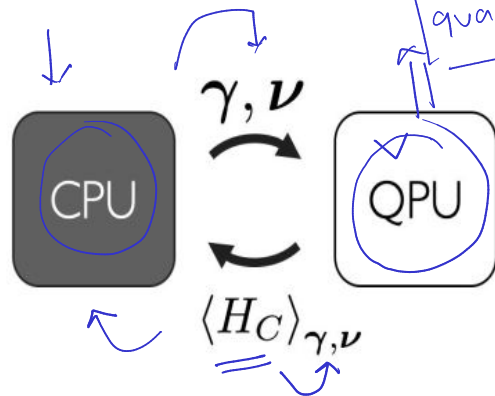
Using quantum processors as accelerators



✓ **REF:** A language and hardware independent approach to quantum–classical computing

Hybrid Strategy II : Hybrid Quantum-Classical Variational Algorithms

Circuit decomposition

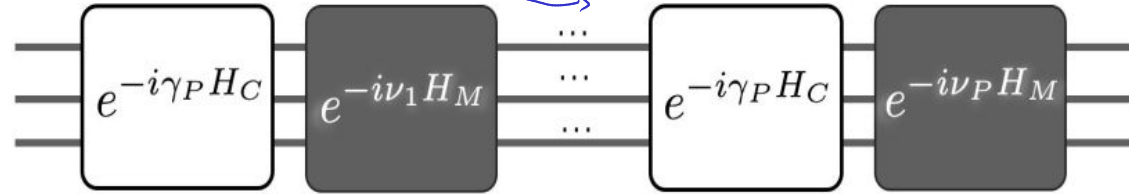


*quantum state
computationally
advantage*

Mixer: $H_M = -\sum_j X_j$

Cost: $H_C = -\sum_{\langle jk \rangle} J_{jk} Z_j Z_k$

variational

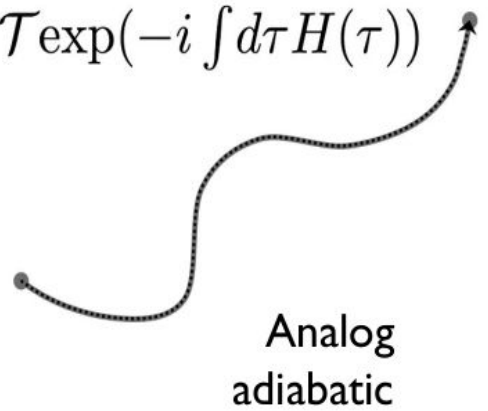


$\langle H \rangle$
 $\langle H \rangle$
 $\langle H \rangle$

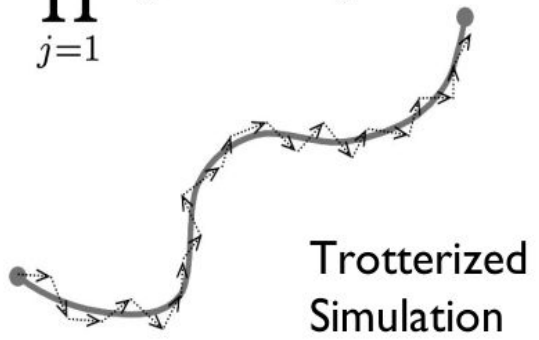
*classical
optimization
of parameters*

$\Rightarrow H(\tau) = (1 - \tau)H_M + \tau H_C$

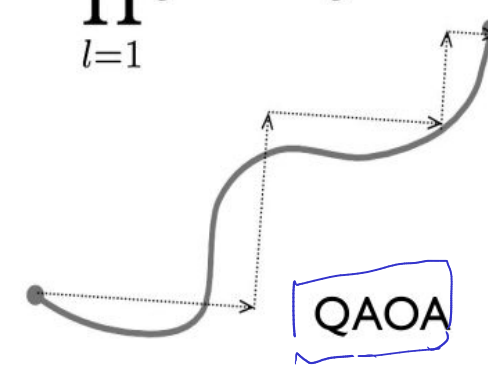
$\Rightarrow \mathcal{T} \exp(-i \int d\tau H(\tau))$



$\prod_{j=1}^{\lceil 1/\Delta\tau \rceil} e^{-i\tau\Delta\tau H_C} e^{-i(1-\tau)\Delta\tau H_M}$



$\prod_{l=1}^P e^{-i\nu_l H_M} e^{-i\gamma_l H_C}$



The quantum and classical worlds

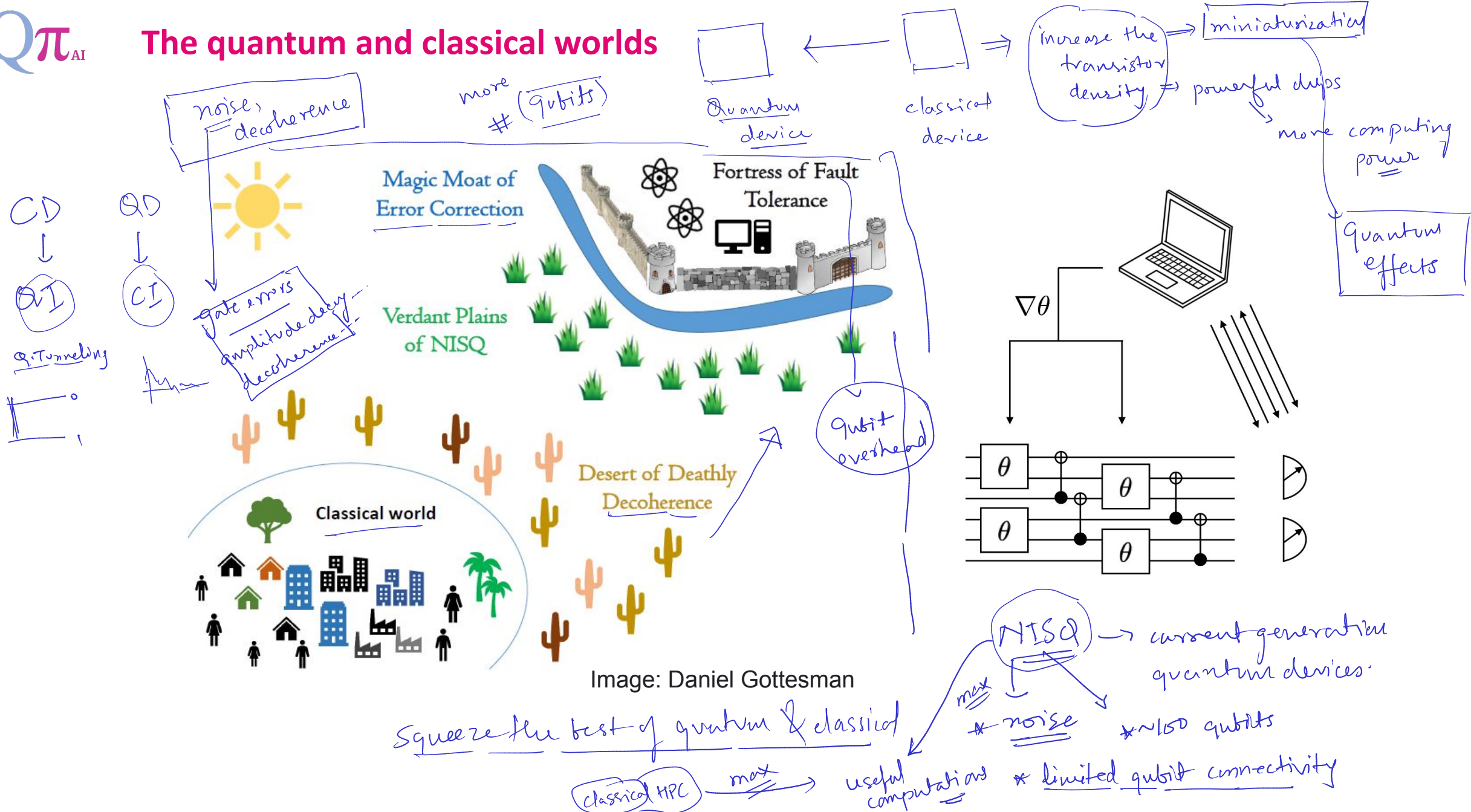


Image: Daniel Gottesman

Squeeze the best of quantum & classical

classical HPC

max

useful computations

NISQ

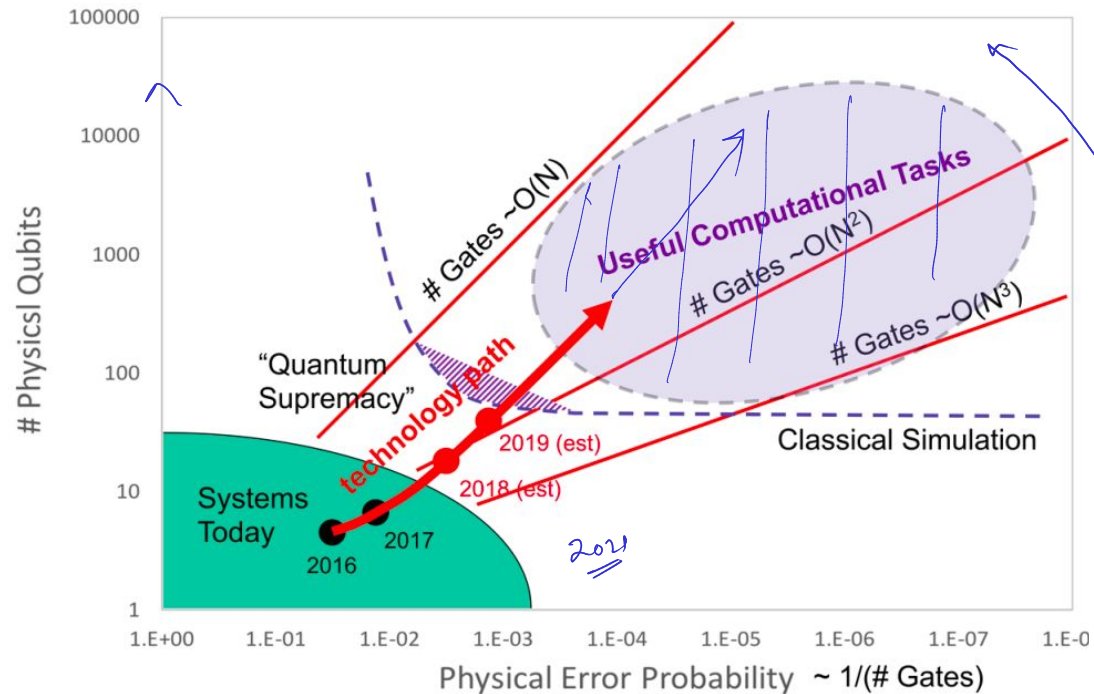
current generation quantum devices

max noise

~150 qubits

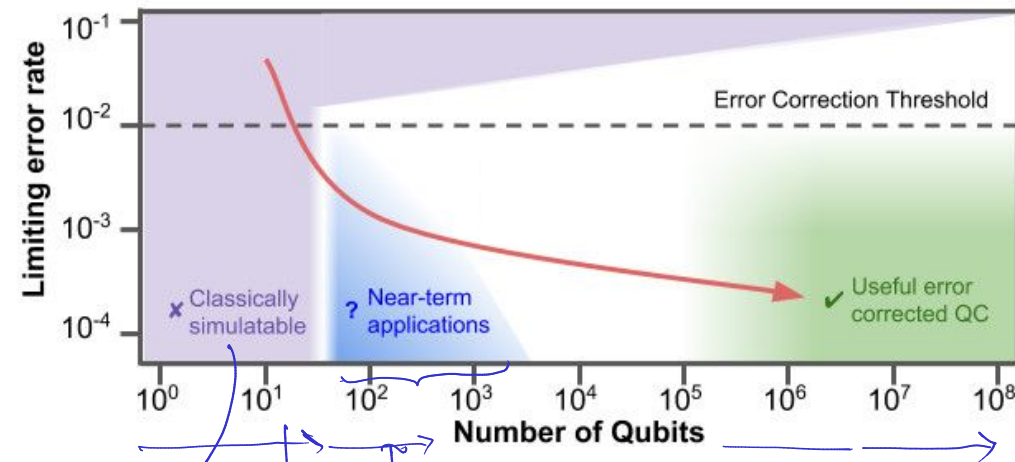
limited qubit connectivity

Noisy, Intermediate Scale Quantum (NISQ) Computers



REF: Next Steps in Quantum Computing: Computer Science's Role, Margaret Martonosi and Martin Roetteler

✓ Suggested reading: Quantum Computing in the NISQ era and beyond, John Preskill, <https://quantum-journal.org/papers/q-2018-08-06-79/>

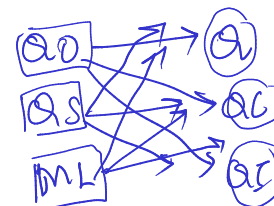


REF: John Martinis

- Quantum optimization
- Quantum simulation
- Machine learning

Modeling optimization tasks as quantum computing

full quantum algorithms
quantum-classical
quantum-inspired



NISQ applications